Purdue University **2026-2027 University Catalog**

Mechanical Engineering, BSME

Location(s): West Lafayette, Indianapolis

About the Program

The BS program in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET, under the commission's General Criteria and Program Criteria for Mechanical and Similarly Named Engineering Programs.

The Bachelor of Science in Mechanical Engineering (BSME) at Purdue University is an ABET-accredited engineering degree that prepares students to become innovators and leaders in the design, analysis, manufacturing, and operation of mechanical systems. Mechanical engineers apply key principles from mathematics, physics, and materials science to solve complex engineering problems and create technologies that improve lives.

Mechanical engineering is one of the oldest and broadest engineering disciplines. Purdue's program emphasizes both fundamental knowledge and practical skills, equipping students to design machines, devices, and systems used throughout modern society. The curriculum builds on foundational coursework in mathematics, physics, and engineering principles to advanced study in core mechanical engineering subjects such as Statics and Dynamics, Thermodynamics and Heat Transfer, Fluid Mechanics, Materials and Manufacturing, and Systems Design and Control.

Students gain hands-on experience through instructional laboratories and design projects that reinforce theoretical concepts while building technical problem-solving skills.

A range of elective courses allows students to tailor their studies to interests such as energy systems, manufacturing, robotics, automotive, and aerospace applications. In senior design, teams solve open-ended, real-world engineering challenges. This course synthesizes technical knowledge and helps students develop engineering design and analysis skills, project management, communication and teamwork, and professional engineering judgment.

Mechanical Engineering students are encouraged to expand their education beyond the classroom through Internships and Co-ops with industry partners, Undergraduate Research with faculty across disciplinary areas, Study Abroad opportunities to gain global engineering perspectives, Student Organizations and Competitions (e.g., engineering clubs, and design teams). These experiences help students build professional skills, explore career paths, and make valuable industry connections.

Graduates with a BSME degree are well-equipped for technical and leadership roles in engineering fields. They go on to work across diverse sectors, including aerospace, automotive, energy, biomedical, robotics, and manufacturing. The solid engineering foundation enables lifelong adaptability and success in rapidly evolving industries.

Mechanical Engineering - Purdue University

Mechanical Engineering Major Change (CODO) Requirements

Degree Requirements

128 Credits Required

Mechanical Engineering Major Requirements (61-62 credits)

A GPA of 2.0 is required in Major Requirements.

- ME 20000 - Thermodynamics I **Credit Hours: 3.00** ♦
- ME 23900 - Introduction To Data Science For Mechanical Engineers **Credit Hours: 1.00** ♦
- ME 26400 - Introduction To Manufacturing For Mechanical Design **Credit Hours: 3.00** ♦
- ME 27000 - Basic Mechanics I **Credit Hours: 3.00** ♦
- ME 27400 - Basic Mechanics II **Credit Hours: 3.00** ♦
- ME 29000 - Global Engineering Professional Seminar (UCC: STS) **Credit Hours: 1.00**
- ME 30800 - Fluid Mechanics **Credit Hours: 3.00** ♦
- ME 30801 - Fluid Mechanics Laboratory **Credit Hours: 1.00** ♦
- ME 31500 - Heat And Mass Transfer **Credit Hours: 4.00** ♦
- ME 32300 - Mechanics Of Materials **Credit Hours: 3.00** ♦
- ME 32301 - Mechanics Of Materials Laboratory **Credit Hours: 1.00** ♦
- ME 35400 - Machine Design **Credit Hours: 3.00** ♦
- ME 36400 - Systematic Engineering Design **Credit Hours: 3.00** ♦
- ME 36500 - Measurement And Control Systems I **Credit Hours: 3.00** ♦
- ME 37500 - Measurement And Control Systems II **Credit Hours: 3.00** ♦
- ME 46300 - Engineering Design **Credit Hours: 3.00** ♦
- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00**
- ECE 20007 - Electrical Engineering Fundamentals I Lab **Credit Hours: 1.00**
- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**
- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- MSE 23000 - Structure And Properties Of Materials **Credit Hours: 3.00**

- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** or
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00**

Other Departmental/Program Course Requirements (35-36 credits)

Other Course Requirements (14-15 credits)

Intro to Engineering I & II - Credit Hours 4.00-8.00

- ENGR 13100 - Transforming Ideas To Innovation I (UCC: AI/IL) **Credit Hours: 2.00** and
- ENGR 13200 - Transforming Ideas To Innovation II (UCC: AI) **Credit Hours: 2.00**

OR

- EPCS 11100 - First Year Participation In EPICS I (UCC: STS) **Credit Hours: 1.00** and
- EPCS 12100 - First Year Participation In EPICS II (UCC: STS) **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: AI/IL) **Credit Hours: 2.00**

OR

- VIP 17911 - First Year Participation In Vertically Integrated Projects (VIP) I (UCC: STS) **Credit Hours: 1.00** and
- VIP 17912 - First Year Participation In Vertically Integrated Projects (VIP) II **Credit Hours: 1.00** and
- ENGR 13300 - Transforming Ideas To Innovation, EPICS/VIP (UCC: IL) **Credits: 2.00**

OR

- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) **Credit Hours: 4.00** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) **Credit Hours: 4.00**

OR

- ENGR 13000 - Transforming Ideas Into Innovations (UCC: AI/IL) **Credit Hours: 4.00**

- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

AND

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

- MFET 16300 - Foundations In Geometric Modeling And Data Management **Credit Hours: 2.00** ♦ or
- ME 16400 - Geometric And Annotation Modeling **Credit Hours: 3.00**

First-Year Engineering Selective - Credit Hours: 3.00-4.00

- CHM 11610 - General Chemistry II **Credit Hours: 3.00**

  AND
- CHM 11620 - General Chemistry II - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11630 - General Chemistry II - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

  OR
- CS 15900 - C Programming **Credit Hours: 3.00**

  OR
- BIOL 11000 - Fundamentals Of Biology I (UCC: SCI) **Credit Hours: 4.00**

  OR
- BIOL 11100 - Fundamentals Of Biology II (UCC: SCI) **Credit Hours: 4.00**

Economics Selective - Credit Hours: 3.00

- ECON 25100 - Microeconomics (UCC: BSS) **Credit Hours: 3.00** or
- ECON 25200 - Macroeconomics (UCC: BSS) **Credit Hours: 3.00**

Physics I Credit Hours: 0.00-4.00

- PHYS 17200 - Modern Mechanics (UCC: SCI) **Credit Hours: 4.00**

  OR
- ENGR 16100 - Honors Introduction To Innovation And The Physical Science Of Engineering Design I (UCC: AI/IL) - **4.00 Credit Hours** and
- ENGR 16200 - Honors Introduction To Innovation And The Physical Science Of Engineering Design II (UCC: SCI) - **4.00 Credit Hours**

- **Technical Selectives** - Credits: 9.00 (see Mechanical Engineering Supplemental Information)
- **Mechanical Engineering Selectives** - Credits: 6.00 (Any ME 300000:59999 that is not already required to fulfill a Major Course requirement)

Engineering General Education Requirements (21 credits)

For a list of courses that satisfy this requirement, see Mechanical Engineering Supplemental Information

- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00
- General Education II - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00

- General Education V - Credit Hours: 3.00
- Written Communication (UCC: WC) - Credit Hours: 3.00-4.00
- Oral Communication (UCC: OC) - Credit Hours: 3.00

Additional General Education Requirements

- *At least 6 credits must be in non-introductory courses (30000-level or above OR from courses with a required pre-requisite in the same department)*
- *At least 9 credits must be taken from the College of Liberal Arts, the Daniels School of Business, and/or the Honors College provided such courses are not focused primarily on engineering, technology, the natural sciences, or mathematics.*
- *A course is not eligible for use towards the 21 credits of Engineering General Education requirement if that course is used to satisfy a different engineering degree requirement.*

Elective (3 credits)

Optional Concentrations

West Lafayette

- Dynamical Systems Modeling & Analysis Concentration for Mechanical Engineering
- Energy Processes and Systems Concentration for Mechanical Engineering
- Manufacturing Concentration for Mechanical Engineering
- Microelectronics and Semiconductors Concentration for Mechanical Engineering

Indianapolis

- Dynamical Systems Modeling & Analysis Concentration for Mechanical Engineering
- Manufacturing Concentration for Mechanical Engineering

Supplemental Information

Click here for Mechanical Engineering Supplemental Information

Non-course/Non-credit Requirements

- At the time of degree completion, a minimum 2.0 GPA in Major Requirements and completion of the ME senior exit survey should be met.

GPA Requirements

- A GPA of 2.0 is required for all Baccalaureate degrees.
- A GPA of 2.0 is required in Major Requirements.

Pass/No Pass Policy

- Courses with a Pass/No Pass grade (P/NP) may not be used to satisfy degree requirements.

Notes

- Students in Mechanical Engineering, BS cannot also enroll in the Aeronautical and Astronautical Engineering, BS or the Motorsports Engineering, BS.

Transfer Credit Policy

If you are interested in registering for a course offered by a different institution, please review the information on the Purdue Office of the Registrar Transfer Credit website.

* If you are an admitted transfer student, please review the above Transfer Credit website and then work with your academic advisor to determine exactly how your previous courses might apply.

University Requirements

University Core Requirements

FOR A COMPLETE LISTING OF UNIVERSITY CORE COURSE SELECTIVES, VISIT THE UNIVERSITY SENATE WEBSITE.

- AI Working Competency (UCC: AI)
- Human Cultures: Behavioral/Social Science (UCC: BSS)
- Human Cultures: Humanities (UCC: HUM)
- Information Literacy (UCC: IL)
- Oral Communication (UCC: OC)
- Quantitative Reasoning (UCC: QR)
- Science #1 (UCC: SCI)
- Science #2 (UCC: SCI)
- Science, Technology, and Society (UCC: STS)
- Written Communication (UCC: WC)

Civics Literacy Proficiency Requirement

The Civics Literacy initiative is designed to develop civic knowledge of Purdue students to graduate a more informed citizenry. For more information visit the Civics Literacy - Office of the Vice Provost for Teaching and Learning - Purdue University.

To obtain the Civics Literacy credential, all Baccalaureate degree seeking students will complete the Civics Literacy Podcast pathway - Complete 12 podcasts created by the Purdue Center for C-SPAN Scholarship & Engagement that use C-SPAN material and pass the required quizzes per episode. After the pathway has been successfully completed, students will take the Civics Literacy knowledge test. Civics Literacy does not impact a student's GPA or bear any financial costs.

More details about each pathway and how to complete the requirement can be found on the Civics Literacy Student Dashboard in myPurdue.

Upper Level Requirement

- Resident study at Purdue University for at least two semesters and the enrollment in and completion of at least 32 semester hours of coursework required and approved for the completion of the degree. These courses are expected to be at least junior-level (30000+) courses.
- Students should be able to fulfill *most*, *if not all*, of these credits within their major requirements; there should be a clear pathway for students to complete any credits not completed within their major.

Sample Plan of Study

The following sample plan of study is a suggested arrangement of courses. Consultation with an advisor may result in an altered plan customized for an individual student.

Fall 1st Year

- Intro to Engineering I – Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- CHM 11510 - General Chemistry I **Credit Hours: 3.00**

- CHM 11520 - General Chemistry I - Laboratory (UCC: SCI) **Credit Hours: 1.00** or
- CHM 11530 - General Chemistry I - Virtual Laboratory (UCC: SCI) **Credit Hours: 1.00**

- MA 16100 - Plane Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 5.00** or
- MA 16500 - Analytic Geometry And Calculus I (UCC: QR) **Credit Hours: 4.00**

13-17 Credits

Spring 1st Year

- Intro to Engineering II - Credit Hours: 2.00-4.00
- Written Communication (UCC: WC) or Oral Communication (UCC:OC) - Credit Hours: 3.00-4.00
- Physics I - Credit Hours: 4.00
- First-Year Engineering Selective - Credit Hours: 3.00-4.00

- MA 16200 - Plane Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 5.00** or
- MA 16600 - Analytic Geometry And Calculus II (UCC: QR) **Credit Hours: 4.00**

16-21 Credits

Fall 2nd Year

- MA 26100 - Multivariate Calculus (UCC: QR) **Credit Hours: 4.00**
- ME 20000 - Thermodynamics I **Credit Hours: 3.00** ♦
- ME 27000 - Basic Mechanics I **Credit Hours: 3.00** ♦
- ME 29000 - Global Engineering Professional Seminar (UCC: STS) **Credit Hours: 1.00**

- MFET 16300 - Foundations In Geometric Modeling And Data Management **Credit Hours: 2.00** ♦ or
- ME 16400 - Geometric And Annotation Modeling **Credit Hours: 3.00** ♦

- PHYS 24100 - Electricity And Optics (UCC: SCI) **Credit Hours: 3.00** or
- PHYS 27200 - Electric And Magnetic Interactions (UCC: SCI) **Credit Hours: 4.00**

16-18 Credits

Spring 2nd Year

- ECE 20001 - Electrical Engineering Fundamentals I **Credit Hours: 3.00** ♦
- ECE 20007 - Electrical Engineering Fundamentals I Lab **Credit Hours: 1.00** ♦
- MA 26600 - Ordinary Differential Equations (UCC: QR) **Credit Hours: 3.00**
- ME 23900 - Introduction To Data Science For Mechanical Engineers **Credit Hours: 1.00**
- ME 26400 - Introduction To Manufacturing For Mechanical Design **Credit Hours: 3.00**
- ME 27400 - Basic Mechanics II **Credit Hours: 3.00**
- General Education I (UCC: HUM) - Credit Hours: 3.00-4.00

17 Credits

Fall 3rd Year

- MA 26500 - Linear Algebra (UCC: QR) **Credit Hours: 3.00**
- ME 30800 - Fluid Mechanics **Credit Hours: 3.00**
- ME 32300 - Mechanics Of Materials **Credit Hours: 3.00**
- ME 32301 - Mechanics Of Materials Laboratory **Credit Hours: 1.00**
- ME 36500 - Measurement And Control Systems I **Credit Hours: 3.00**

- ECON 25100 - Microeconomics (UCC: BSS) **Credit Hours: 3.00** or
- ECON 25200 - Macroeconomics (UCC: BSS) **Credit Hours: 3.00**

16 Credits

Spring 3rd Year

- ME 30801 - Fluid Mechanics Laboratory **Credit Hours: 1.00**
- ME 35400 - Machine Design **Credit Hours: 3.00**
- ME 36400 - Systematic Engineering Design **Credit Hours: 3.00**
- ME 37500 - Measurement And Control Systems II **Credit Hours: 3.00**
- Technical Elective I - Credit Hours: 3.00
- General Education II (UCC: BSS) - Credit Hours: 3.00

16 Credits

Fall 4th Year

- ME 31500 - Heat And Mass Transfer **Credit Hours: 4.00**
- MSE 23000 - Structure And Properties Of Materials **Credit Hours: 3.00**
- Mechanical Engineering Selective I - Credit Hours: 3.00
- Technical Selective II - Credit Hours: 3.00
- General Education III - Credit Hours: 3.00

16 Credits

Spring 4th Year

- ME 46300 - Engineering Design **Credit Hours: 3.00**
- Mechanical Engineering Selective II - Credit Hours: 3.00
- Technical Selective III - Credit Hours: 3.00
- General Education IV - Credit Hours: 3.00
- General Education V – Credit Hours: 3.00
- Elective - Credit Hours: 3.00

18 Credits

Pre-Requisite Information

For pre-requisite information, log in to mypurdue.purdue.edu and click here.

World Language Courses

World Language proficiency requirements vary by program. The following list is inclusive of all world languages PWL offers for credit; for acceptable languages and proficiency levels, see your advisor. (ASL-American Sign Language; ARAB-Arabic; CHNS-Chinese; FR-French; GER-German; GREK-Greek(Ancient); HEBR-Hebrew(Biblical); HEBR-Hebrew(Modern); ITAL-Italian; JPNS-Japanese; KOR-Korean; LATN-Latin; PTGS-Portuguese; RUSS-Russian; SPAN-Spanish)

Critical Course

The ♦ course is considered critical.

In alignment with the Degree Map Guidance for Indiana's Public Colleges and Universities, published by the Commission for Higher Education (pursuant to HEA 1348-2013), a Critical Course is identified as "one that a student must be able to pass to persist and succeed in a particular major. Students who want to be nurses, for example, should know that they are expected to be proficient in courses like biology in order to be successful. These would be identified by the institutions for each degree program."

Disclaimer

The student is ultimately responsible for knowing and completing all degree requirements. Consultation with an advisor may result in an altered plan customized for an individual student. The myPurduePlan powered by DegreeWorks is the knowledge source for specific requirements and completion.

Comparative information about Purdue University and other U.S. educational institutions is also available through the College Navigator tool, provided by the National Center for Education Statistics, and through the U.S. Department of Education College Scorecard.